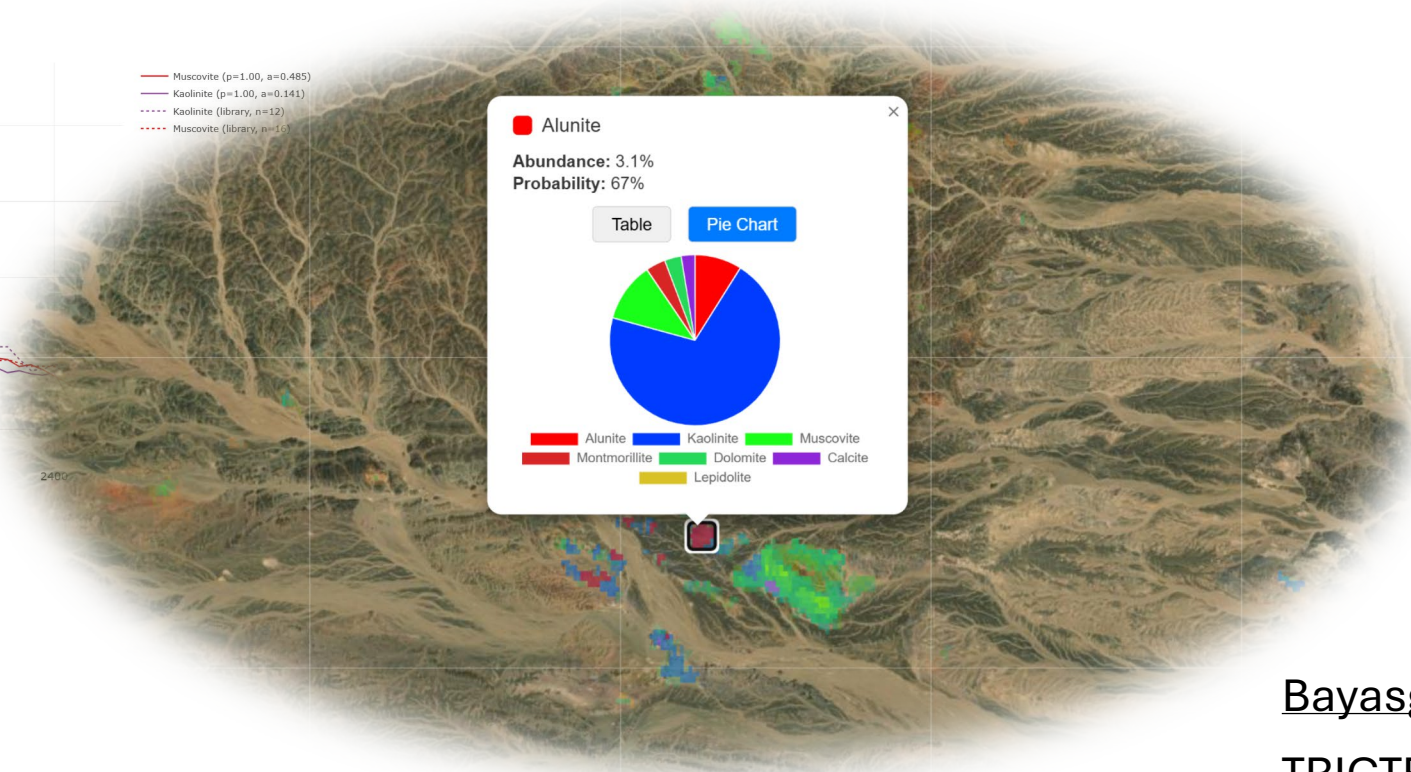
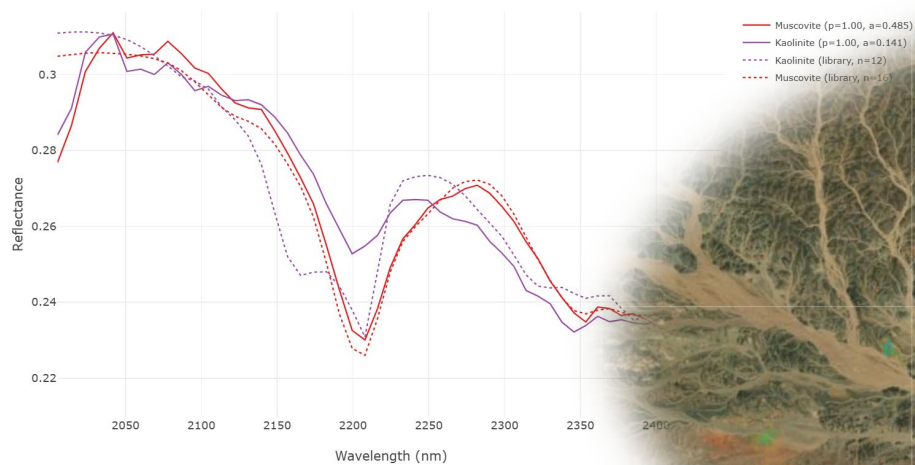


DETERMINATION OF ALTERATION ZONING IN PORPHYRY TYPE DEPOSITS USING HYPERSPECTRAL IMAGERY



Bill Basener, PhD

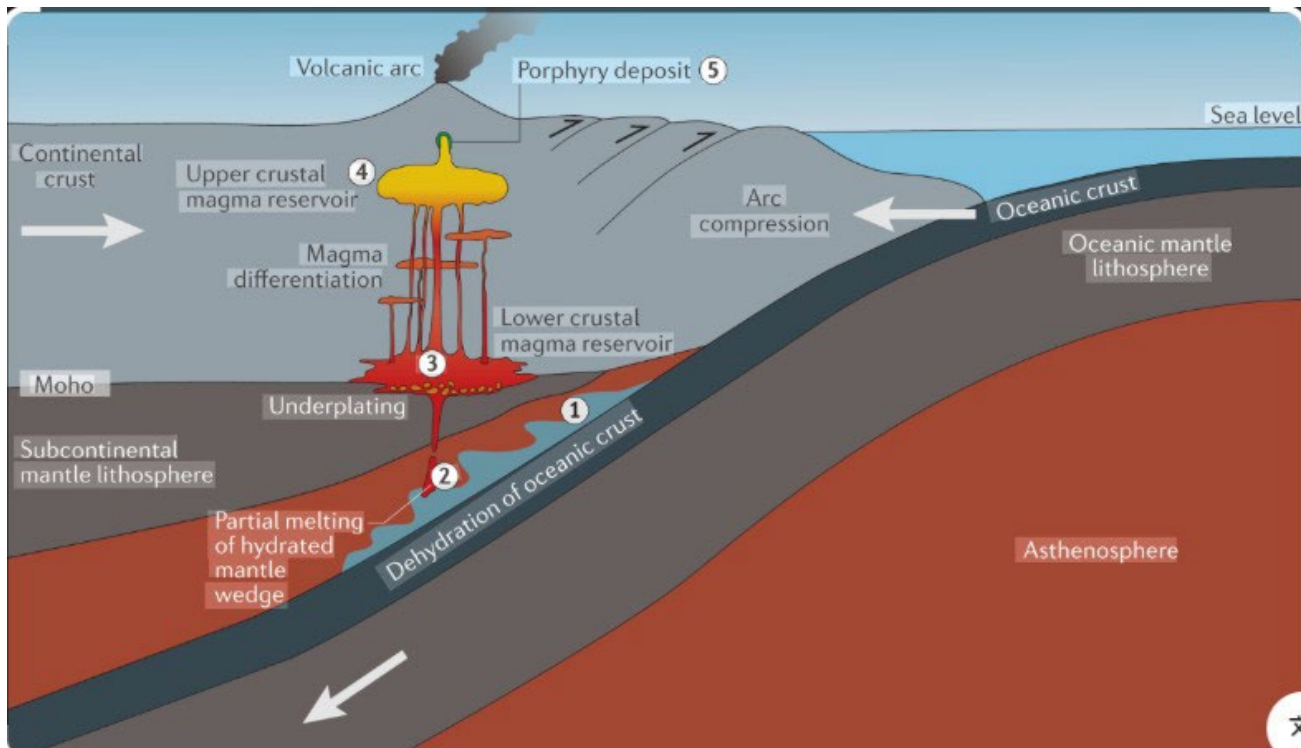
Geospatial Technology Associates

GTA

Bayasgalan Amgalan, PhD

TRIGTEQ LLC

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Porphyry deposits are large, low- to medium-grade mineral deposits associated with shallow, felsic intrusive rocks. They are the world's primary source of copper and molybdenum, and major providers of gold, silver, and other byproducts. Because of their massive tonnage, they are highly economical to mine using bulk extraction methods.

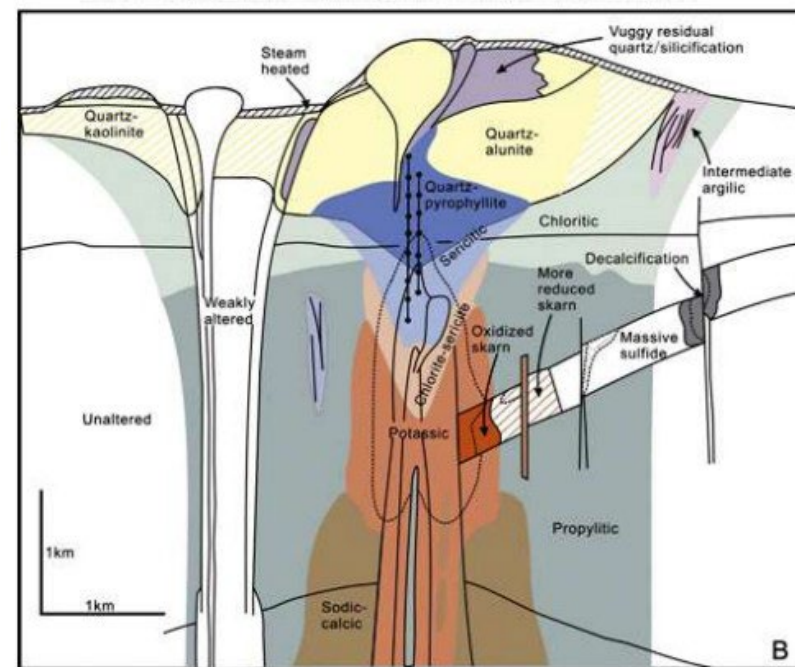
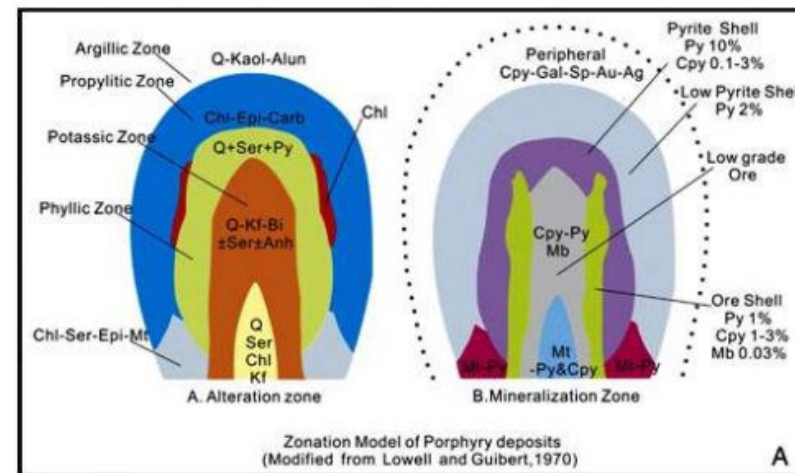
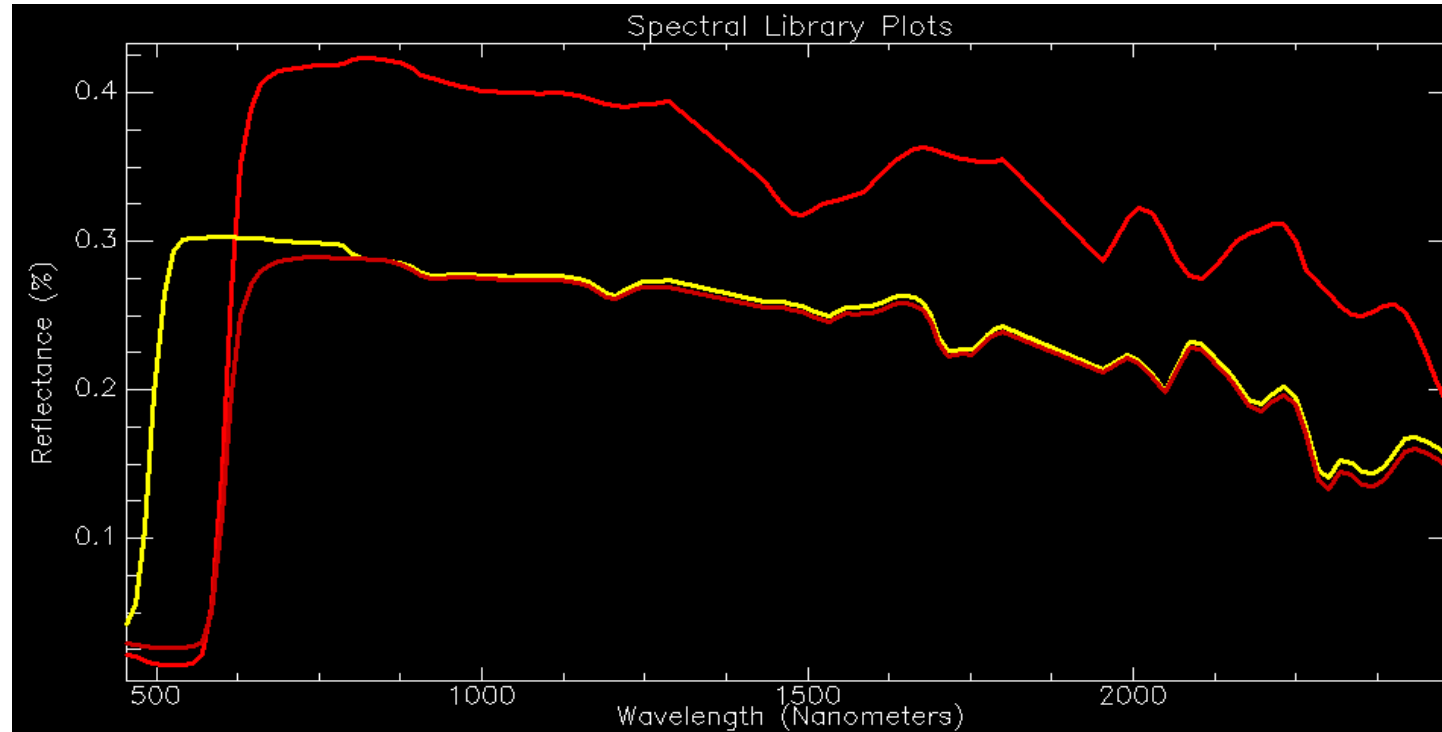


TABLE 2. Characteristics of Principal Alteration-Mineralization Types in Porphyry Cu Systems¹

Alteration type ² (alternative name)	Position in system (abundance)	Key minerals	Possible ancillary minerals	Principal sulfide assemblages (minor)	Contemporaneous veinlets ³ (designation)	Veinlet selvages	Economic potential
Sodic-calcic	Deep, including below porphyry Cu deposits (uncommon)	Albite/oligoclase, actinolite, magnetite	Diopside, epidote, garnet	Typically absent	Magnetite ± actinolite (M-type)	Albite/oligoclase	Normally barren, but locally ore bearing
Potassic (K-silicate)	Core zones of porphyry Cu deposits (ubiquitous)	Biotite, K-feldspar	Actinolite, epidote, sericite, andalusite, albite, carbonate, tourmaline, magnetite	Pyrite-chalcopyrite, chalcopyrite ± bornite, bornite ± digenite ± chalcocite	Biotite (EB-type), K-feldspar, quartz-biotite-sericite- K-feldspar-andalusite- sulfides (EDM/T4-type), quartz-sulfides ± magnetite (A-type), quartz-molybdenite ± pyrite ± chalcopyrite (central suture; B-type)	EDM-type with sericite ± biotite ± K-feldspar ± andalusite + disseminated chalcopyrite ± bornite; others none, except locally K-feldspar around A- and B-types	Main ore contributor
Propylitic	Marginal parts of systems, below lithocaps (ubiquitous)	Chlorite, epidote, albite, carbonate	Actinolite, hematite, magnetite	Pyrite (± sphalerite, galena)	Pyrite, epidote		Barren, except for subepither- mal veins
Chlorite-sericite (sericite-clay-chlorite [SCC])	Upper parts of porphyry Cu core zones (common, particularly in Au- rich deposits)	Chlorite, sericite/illite, hematite (martite, specularite)	Carbonate, epidote, smectite	Pyrite-chalcopyrite	Chlorite ± sericite ± sulfides	Chlorite, sericite/illite	Common ore contributor
Sericitic (phyllic)	Upper parts of porphyry Cu deposits (ubiquitous, except with alkaline intrusions)	Quartz, sericite	Pyrophyllite, carbonate, tourmaline, specularite	Pyrite ± chalcopyrite (pyrite-enargite ± tennantite, pyrite- bornite ± chalcocite, pyrite-sphalerite)	Quartz-pyrite ± other sulfides (D-type)	Quartz-sericite	Commonly barren, but may constitute ore
Advanced argillic (secondary quartzite in Russian terminology)	Above porphyry Cu deposits, constitutes lithocaps (common)	Quartz (partly residual, vuggy), alunite, ⁴ pyrophyllite, dickite, kaolinite	Diaspore, andalusite, zunyite, corundum, dumortierite, topaz, specularite	Pyrite-enargite, pyrite-chalcocite, pyrite-covellite	Pyrite-enargite ± Cu sulfides (includes veins)	Quartz-alunite, quartz- pyrophyllite/dickite, quartz-kaolinite	Locally constitutes ore in lithocaps and their roots

¹ Excluding those developed in carbonate-rich rocks² Arranged from probable oldest (top) to youngest (bottom), except for propylitic that is lateral equivalent of potassic; advanced argillic also forms above potassic early in systems (Fig. 10)³ Many veinlets in potassic, chlorite-sericite, and sericitic alteration contain anhydrite, which also occurs as late, largely monomineralic veinlets⁴ Alunite commonly intergrown with aluminum-phosphate-sulfate (APS) minerals (see Stoffregen and Alpers, 1987)

Spectroscopy

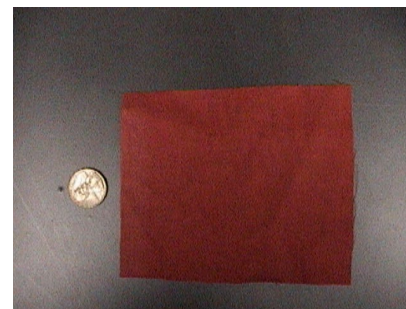


Visible light reflectance determined by photons causing electrons to move among levels.

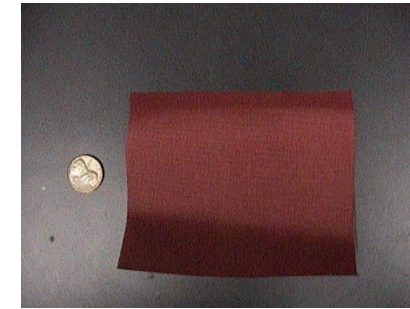
Near Infrared to Short Wave Infrared reflectance determined by photons causing vibrations in bonds.



Yellow Nylon



Red Cotton



Red Nylon

+

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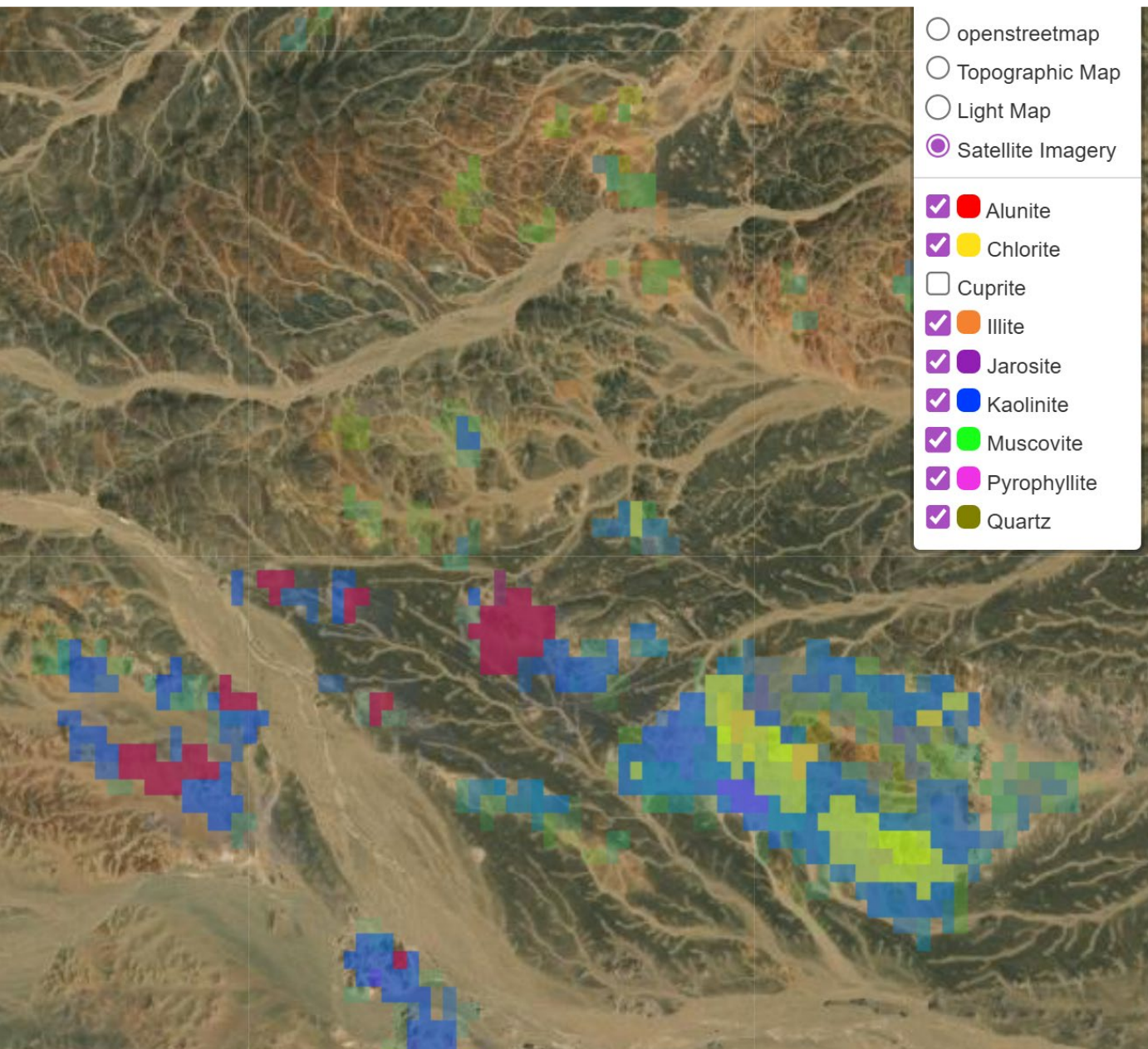
OreVision

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- ☒ Cuprite
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- ☒ Jarosite
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- ☒ Muscovite
- ☒ Pyrophyllite
- ☒ Quartz

GTA

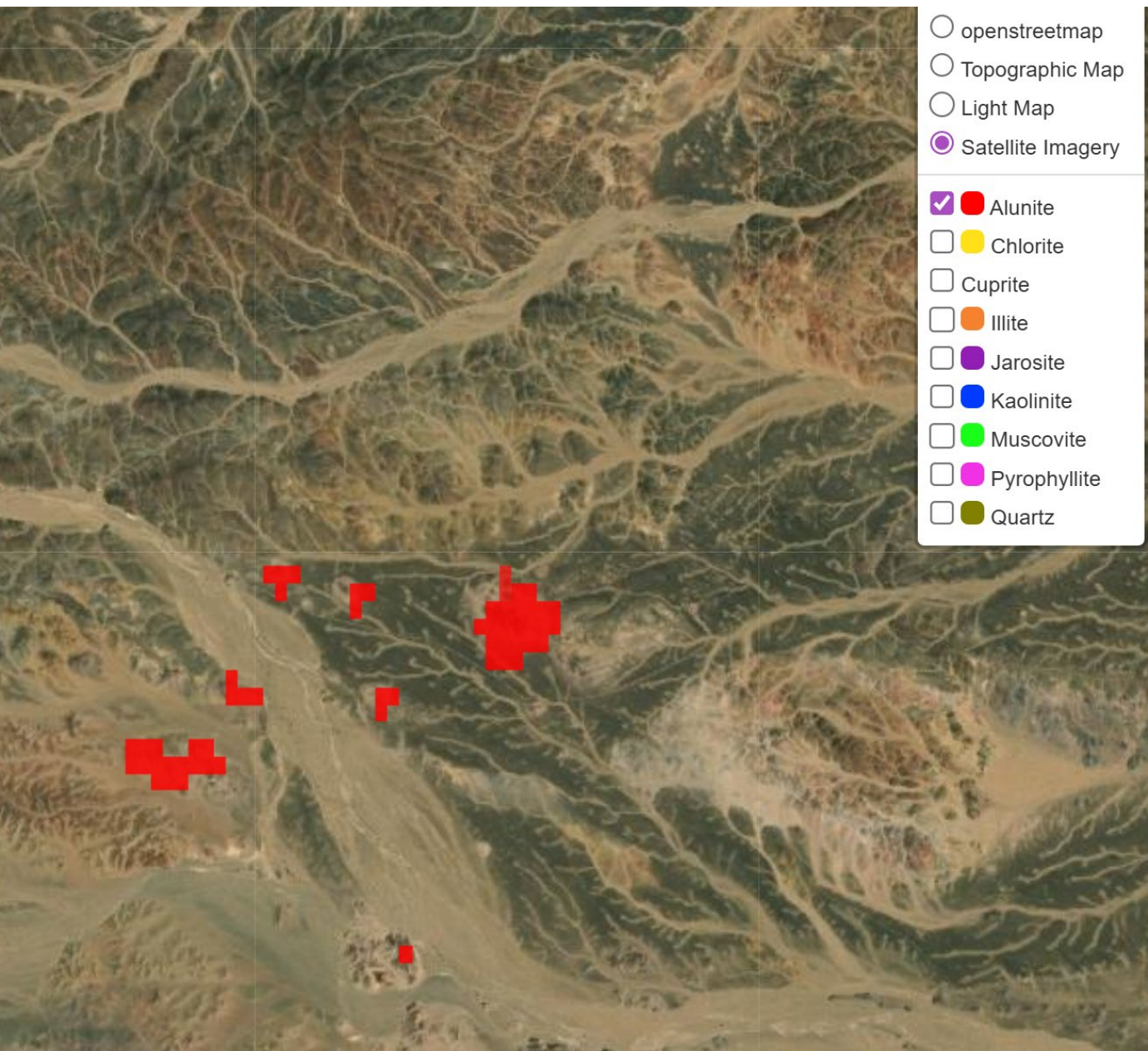
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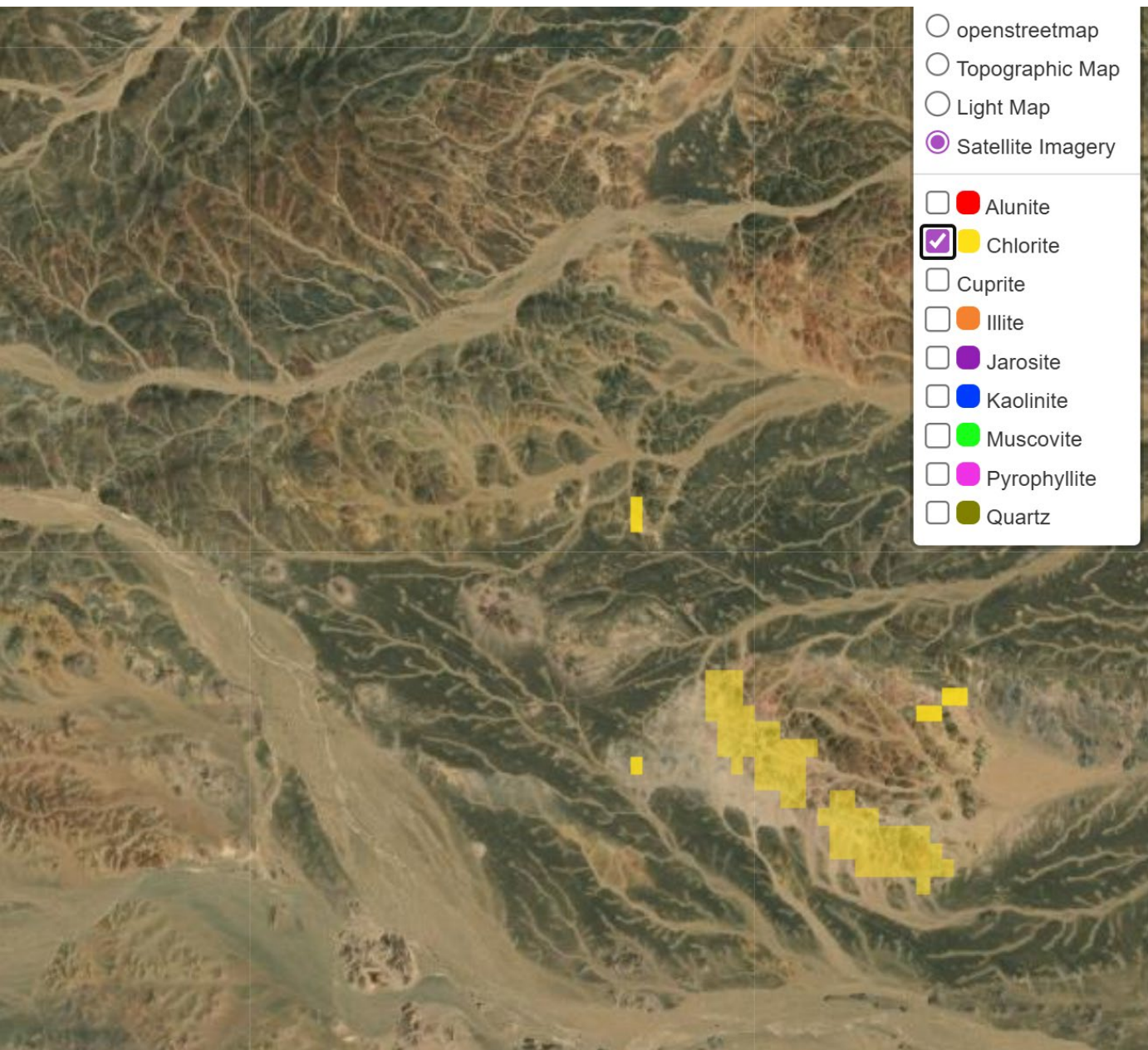
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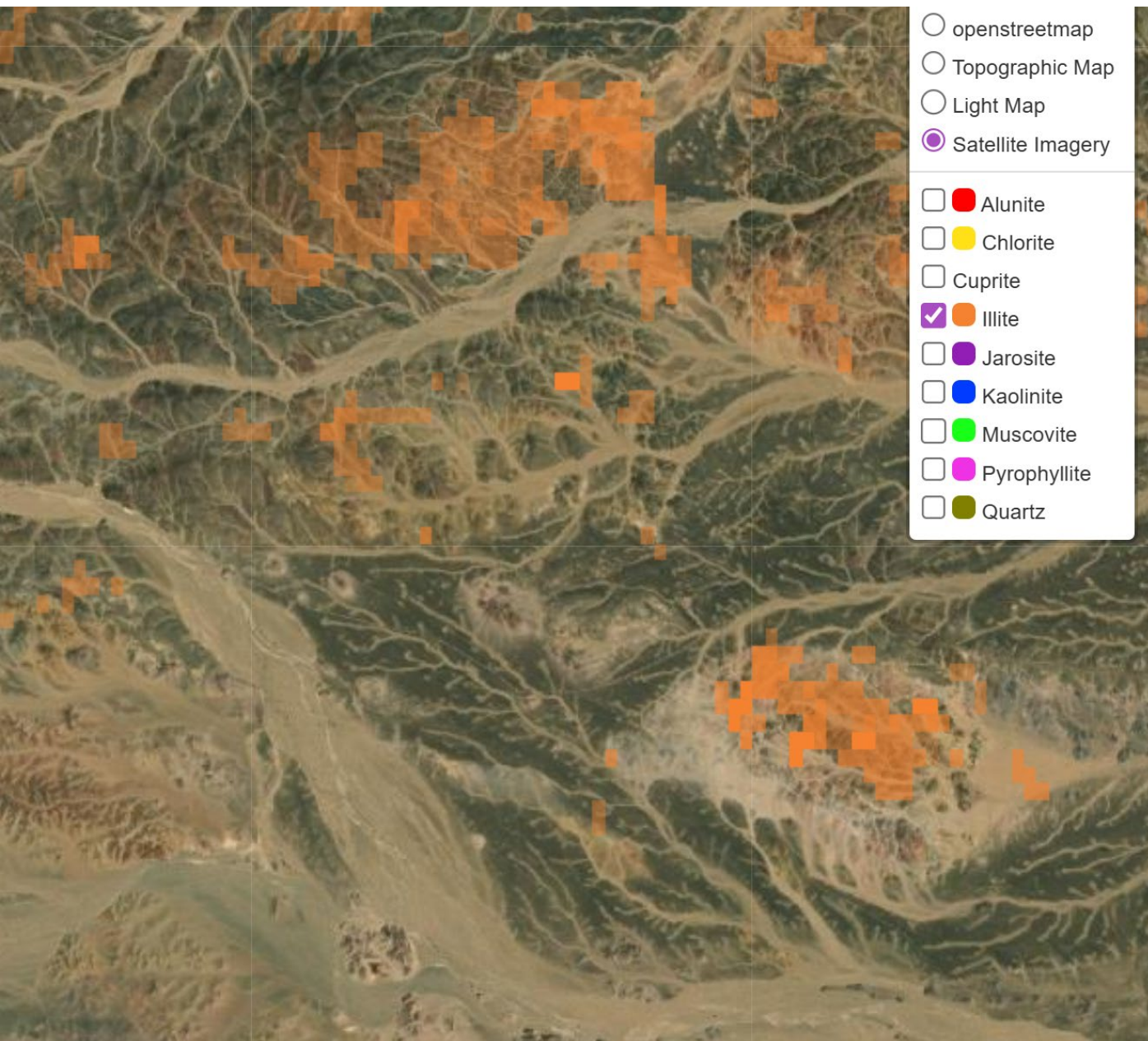
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Potassic (K-silicate) Main ore contributor	Core zones of porphyry Cu deposits (ubiquitous)	Biotite, K-feldspar	Actinolite, epidote, sericite, andalusite, albite, carbonate, tourmaline, magnetite
Propylitic Barren, except for subepither- mal veins	Marginal parts of systems, below lithocaps (ubiquitous)	Chlorite, epidote, albite, carbonate	Actinolite, hematite, magnetite
Chlorite-sericite (sericite-clay-chlorite [S]) Common ore contributor	Upper parts of porphyry Cu core zones (common, particularly in Au- rich deposits)	Chlorite, sericite/illite, hematite (martite, specularite)	Carbonate, epidote, smectite
Sericitic (phyllic) Commonly barren, but may constitute ore	Upper parts of porphyry Cu deposits (ubiquitous, except with alkaline intrusions)	Quartz, sericite	Pyrophyllite, carbonate, tourmaline, specularite
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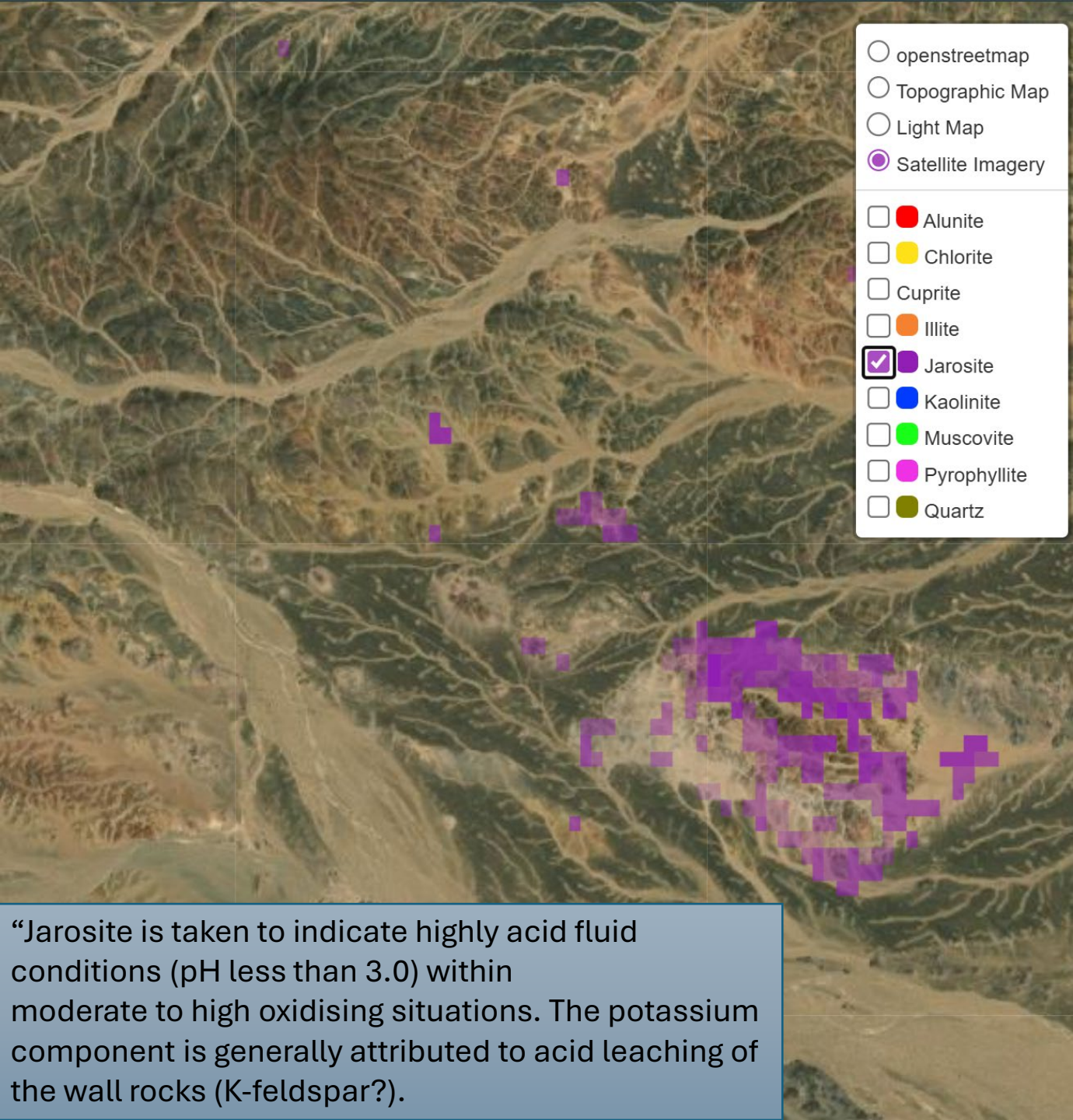
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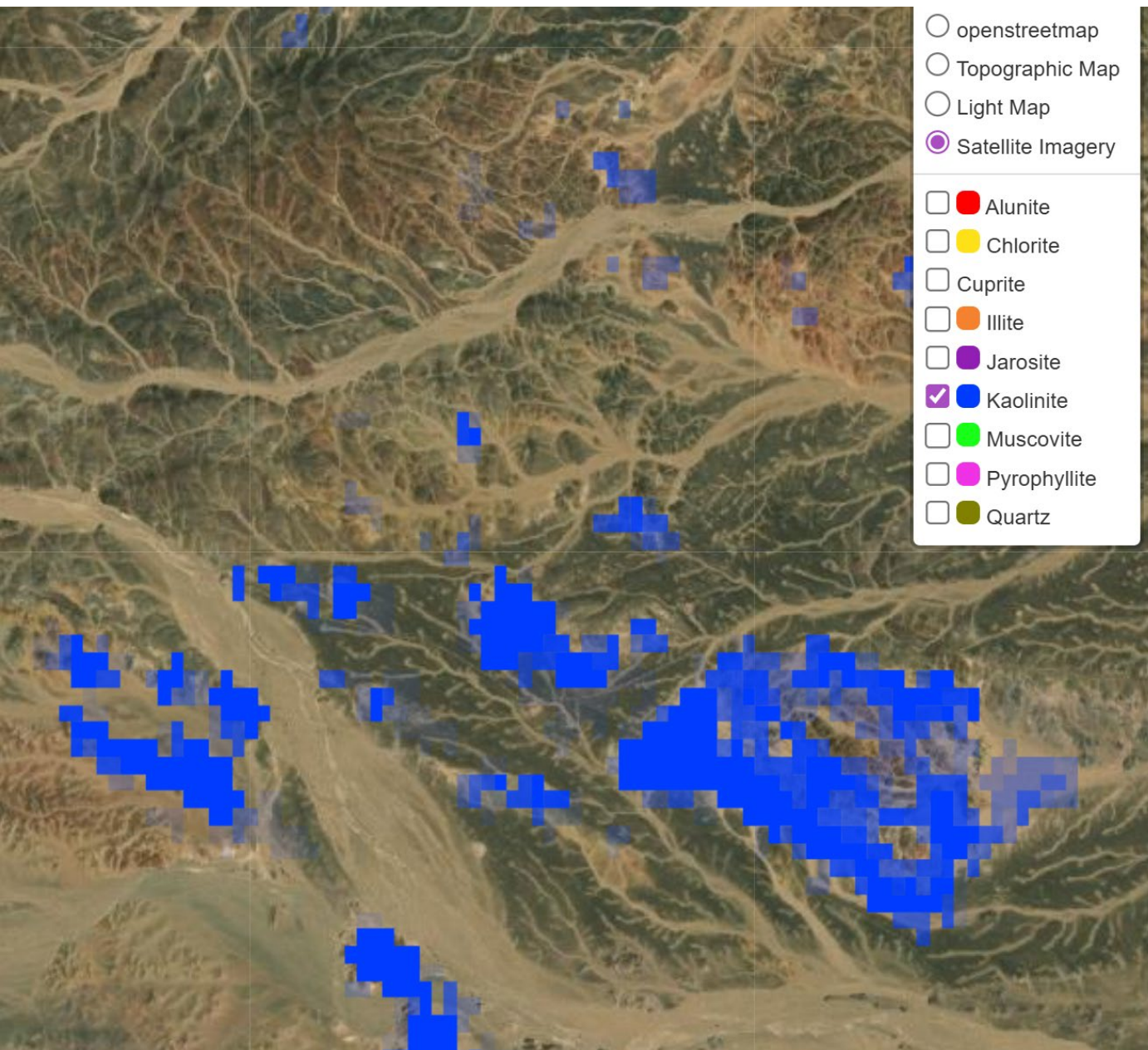
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“Jarosite is taken to indicate highly acid fluid conditions (pH less than 3.0) within moderate to high oxidising situations. The potassium component is generally attributed to acid leaching of the wall rocks (K-feldspar?).

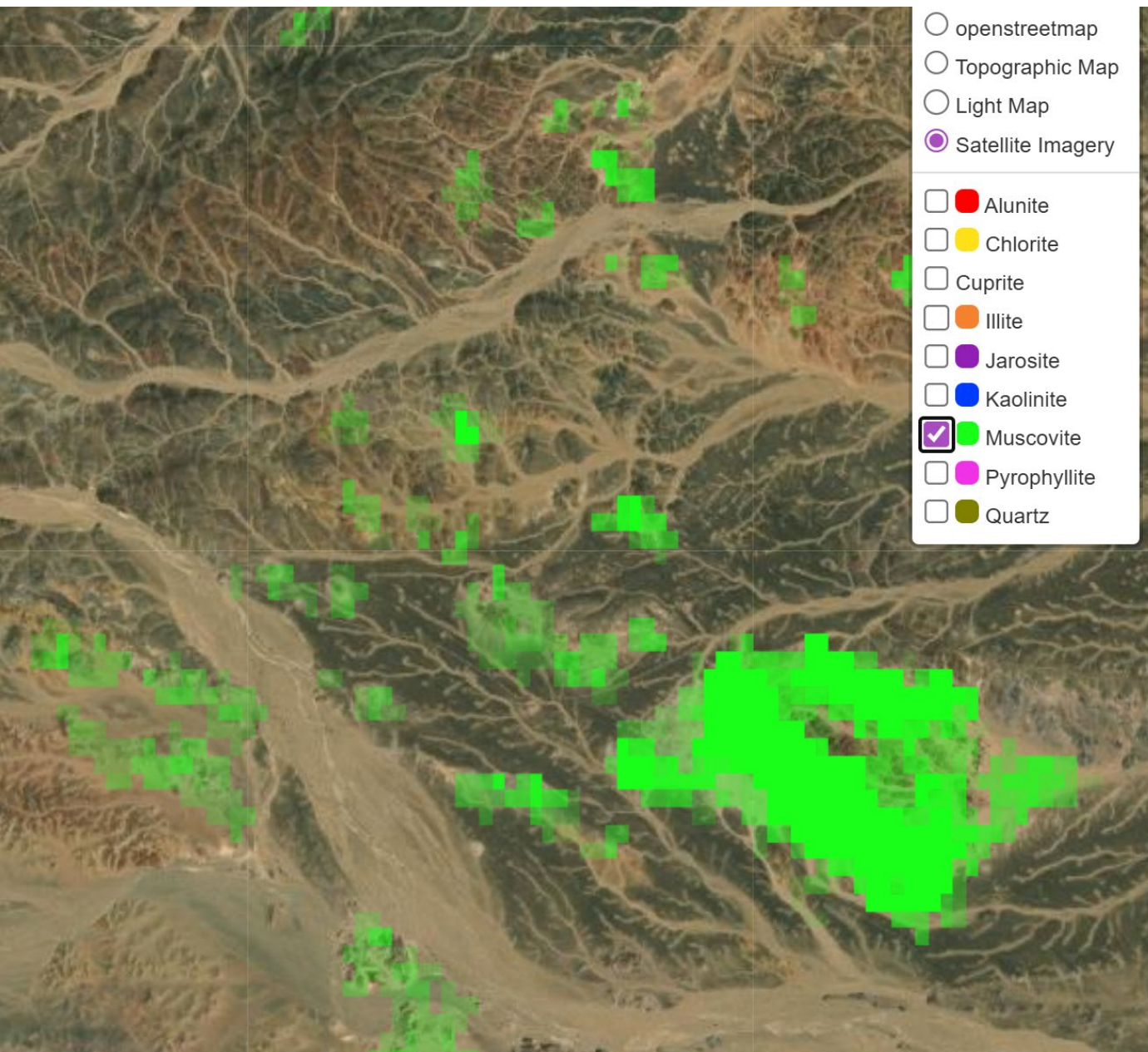
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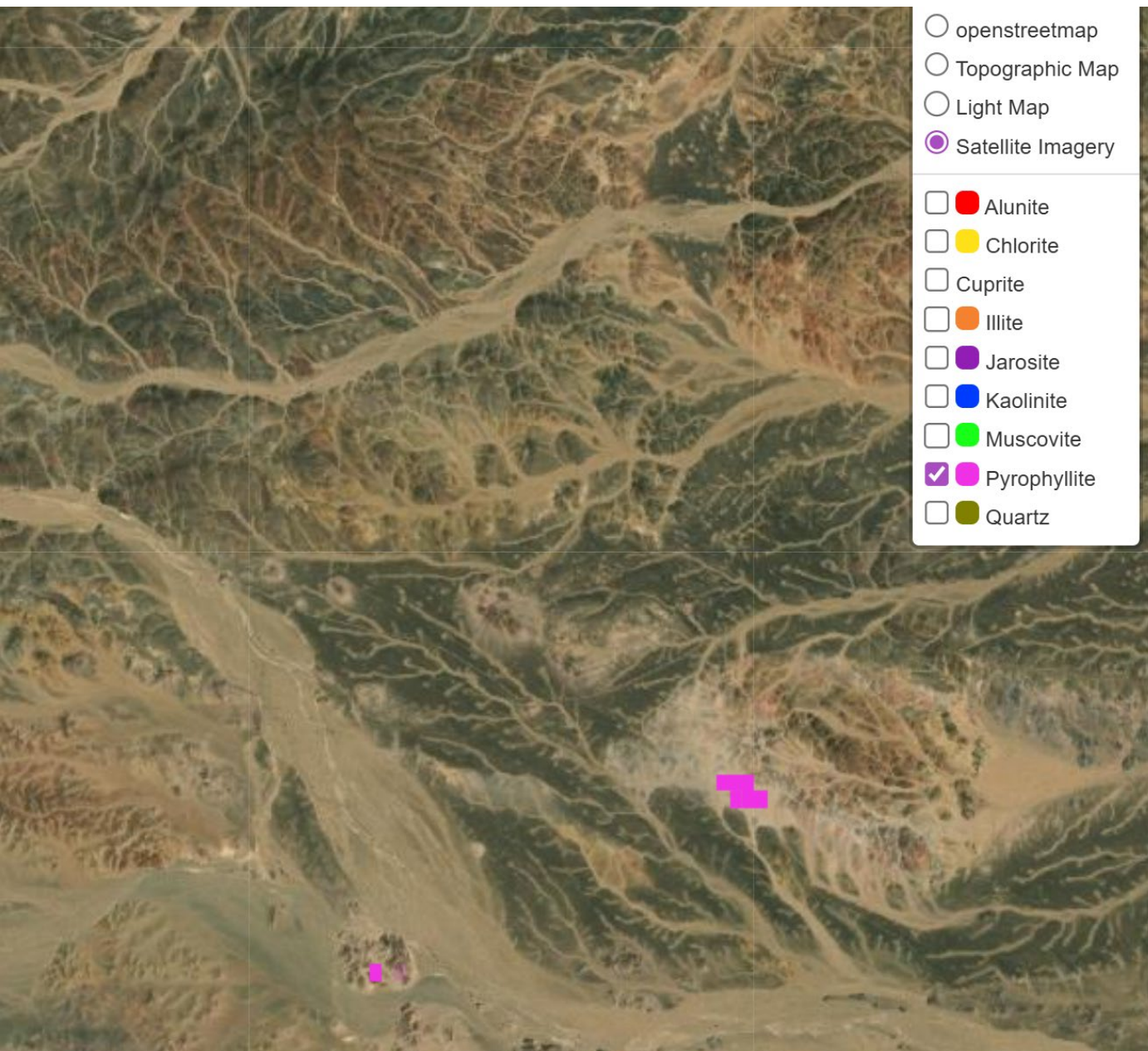
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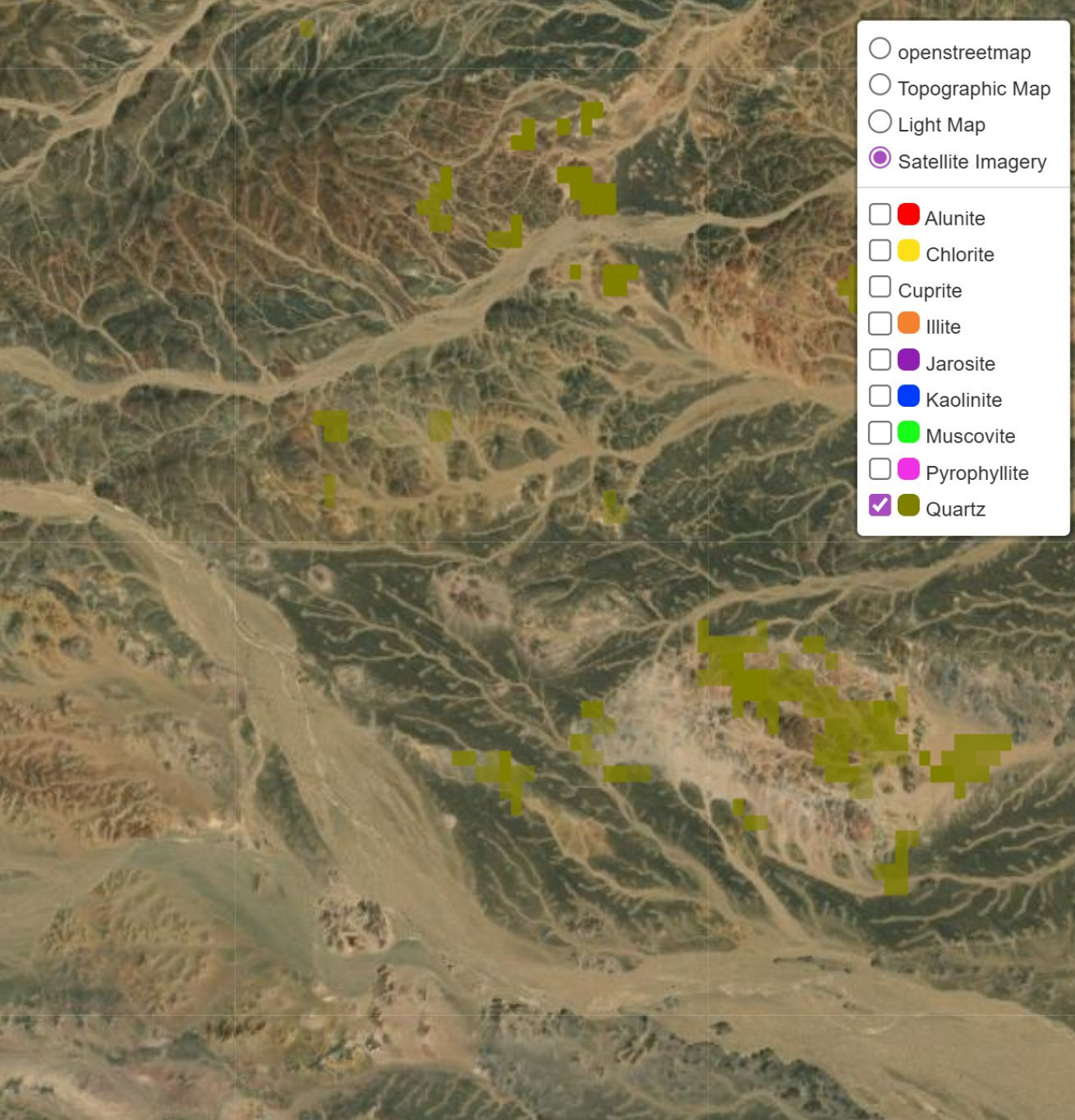
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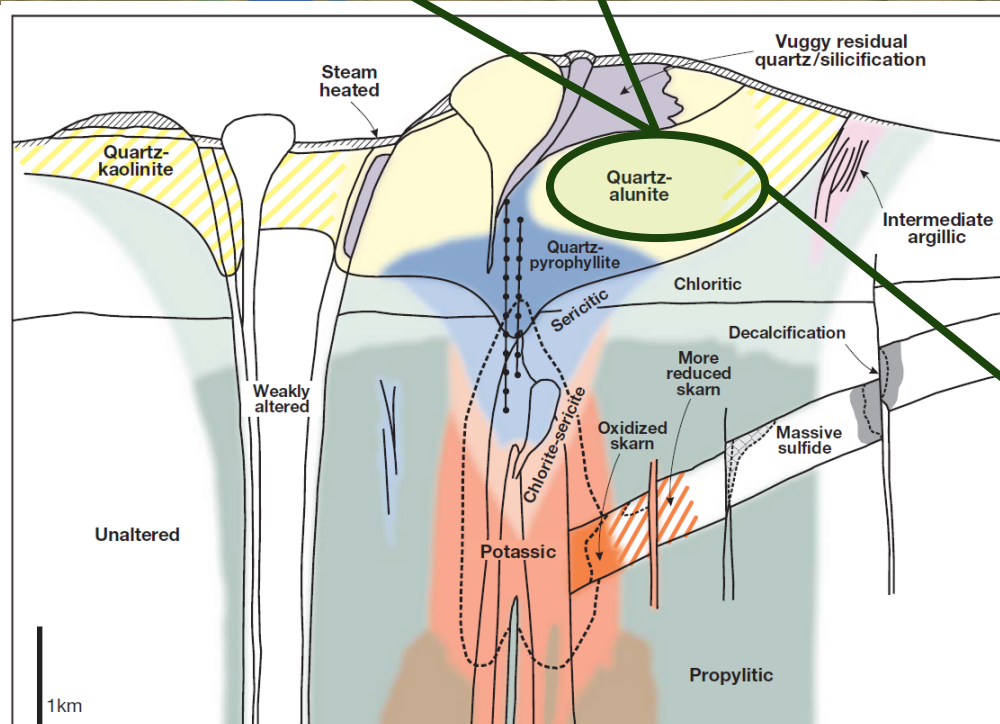
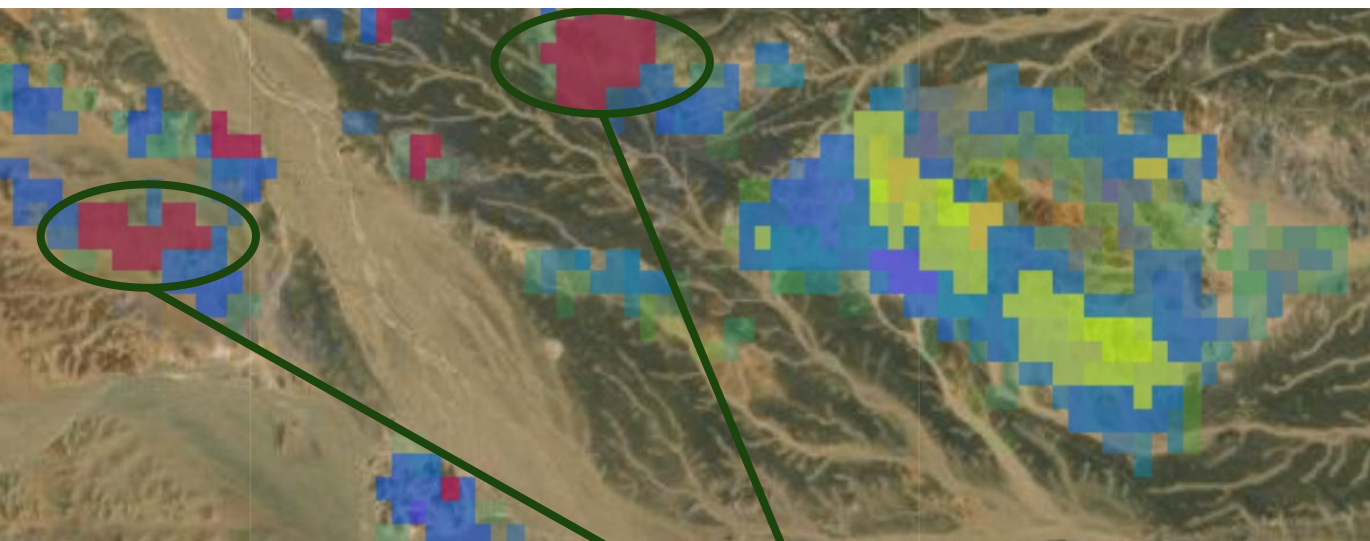
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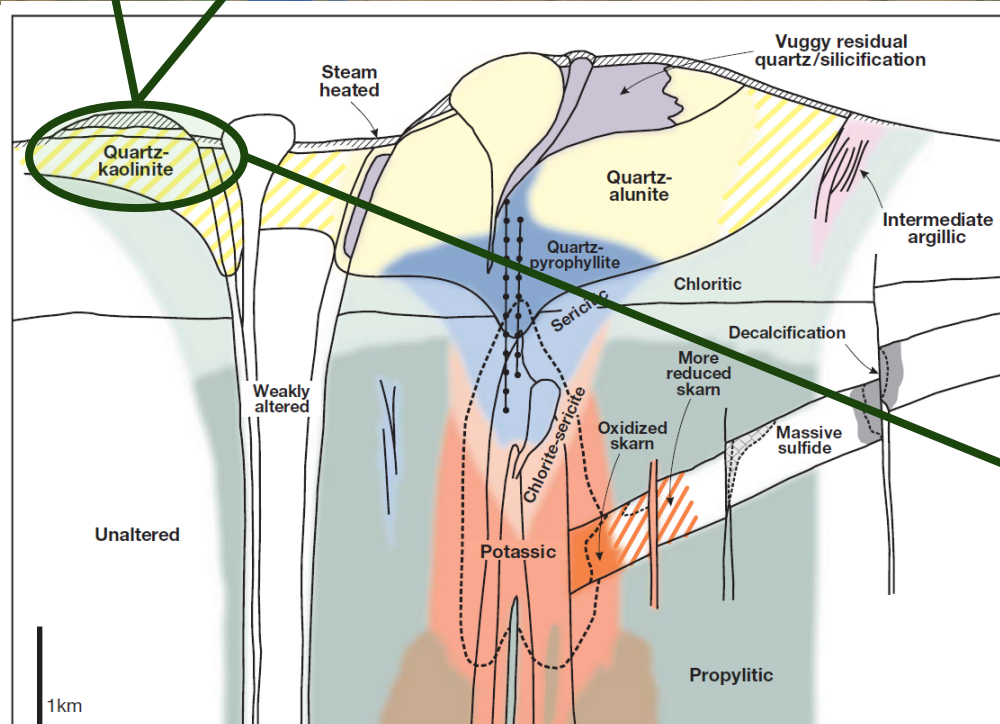
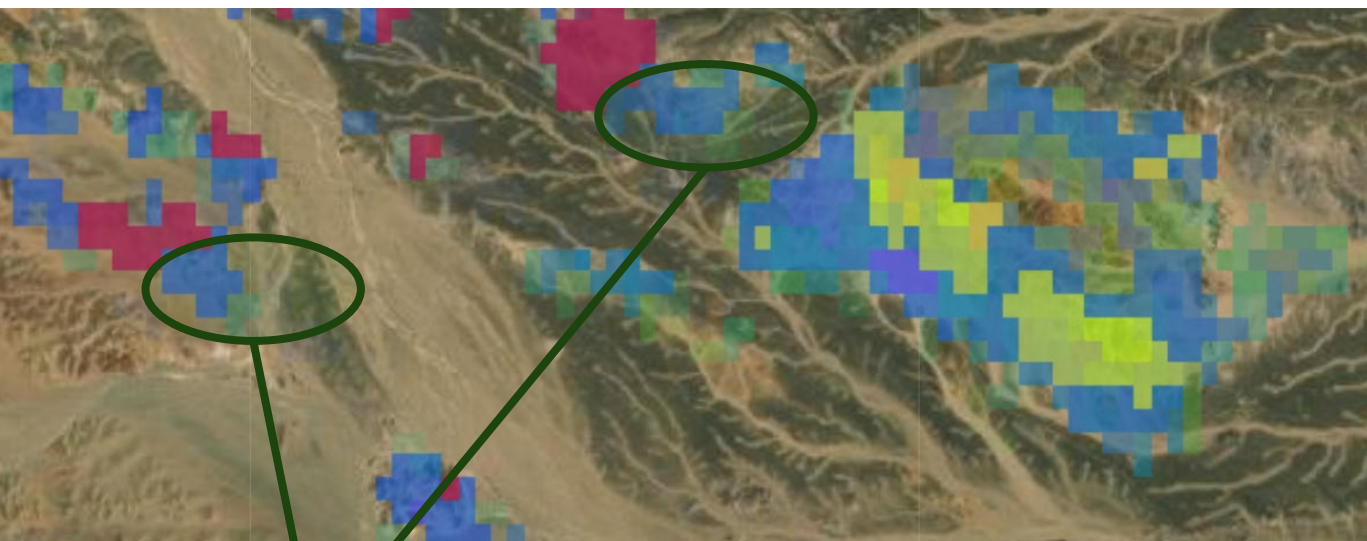
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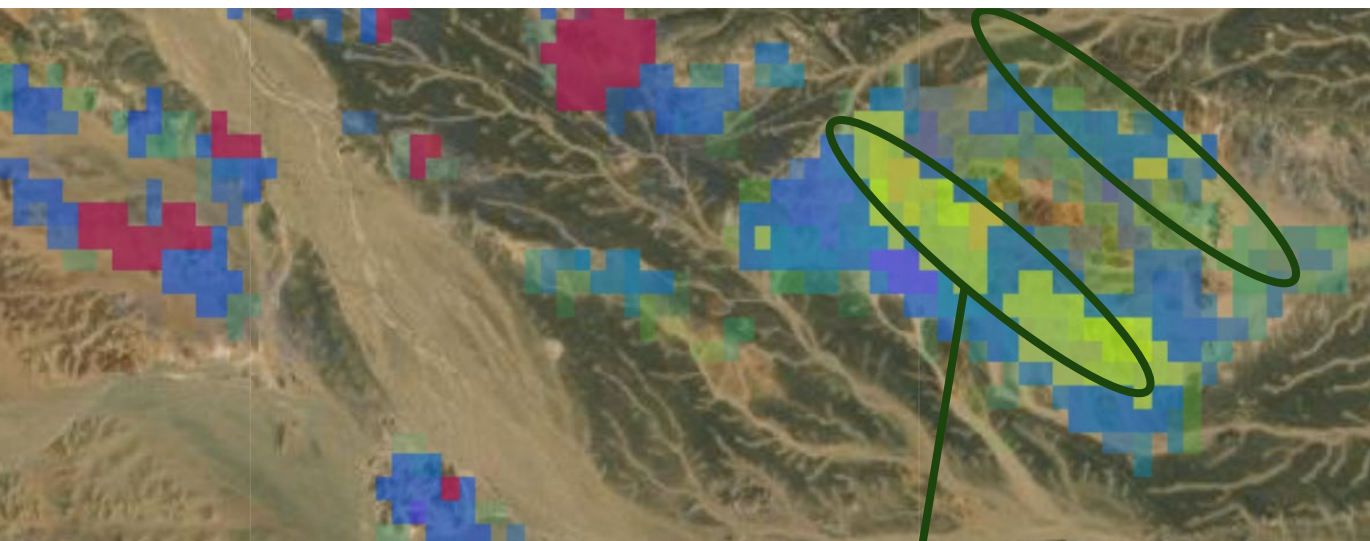
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Propylitic Barren, except for subepither- mal veins	Marginal parts of systems, below lithocaps (ubiquitous)	Chlorite, epidote, albite, carbonate	Actinolite, hematite, magnetite
Chlorite-sericite (sericite-clay-chlorite) Common ore contributor	Upper parts of porphyry Cu core zones (common, particularly in Au- rich deposits)	Chlorite, sericite/illite, hematite (martite, specularite)	Carbonate, epidote, smectite
Sericitic (phyllic) Commonly barren, but may constitute ore	Upper parts of porphyry Cu deposits (ubiquitous, except with alkaline intrusions)	Quartz-sericite Pyrophyllite	carbonate, tourmaline, specularite
Advanced argillic (secondary quartzite in Russian terminology) Locally constitutes ore in lithocaps and their roots	Above porphyry Cu deposits, constitutes lithocaps (common)	Quartz (partly residual, vuggy), alunite, ⁴ pyrophyllite,	Diaspore, andalusite, zunyite, corundum, dumortierite, topaz, specularite



Alteration type ² (alternative name)	Position in system (abundance)	Key minerals	Possible ancillary minerals
Sodic-calcic Normally barren, but locally ore bearing	Deep, including below porphyry Cu deposits (uncommon)	Albite/oligoclase, actinolite, magnetite	Diopside, epidote, garnet
Potassic (K-silicate) Main ore contributor	Core zones of porphyry Cu deposits (ubiquitous)	Biotite, K-feldspar	Actinolite, epidote, sericite, andalusite, albite, carbonate, tourmaline, magnetite

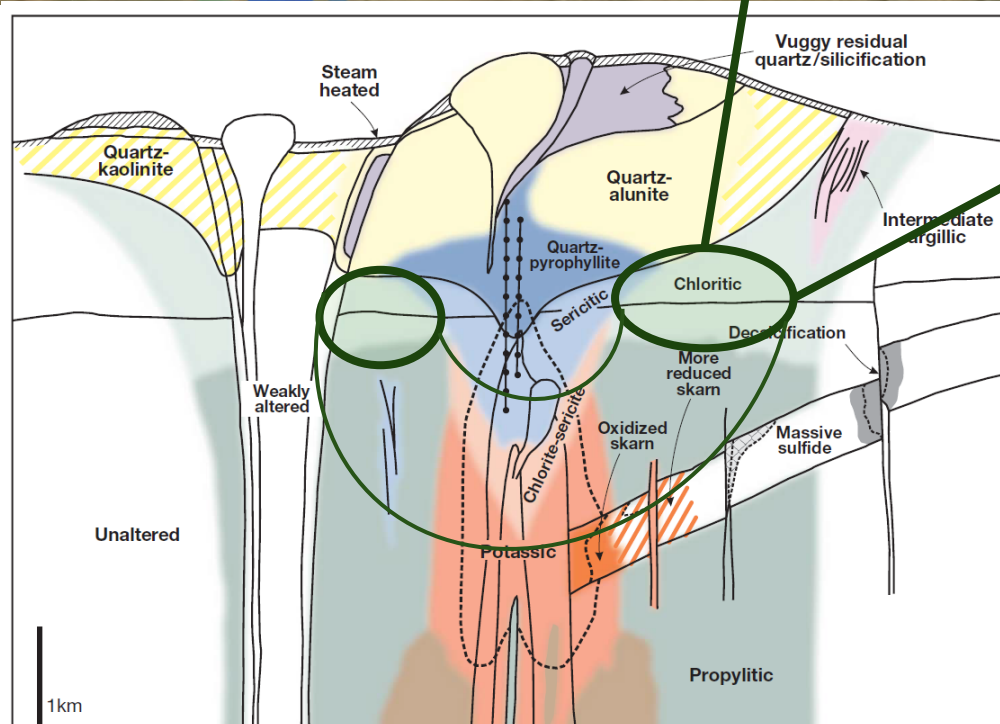
Annulus around deposit, Distal

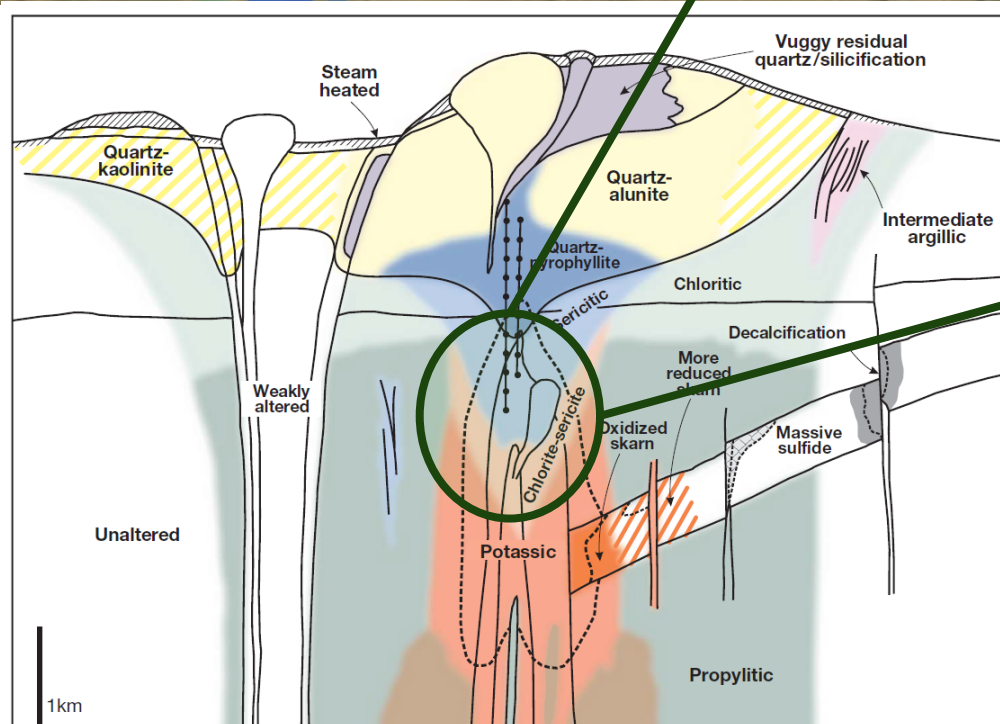
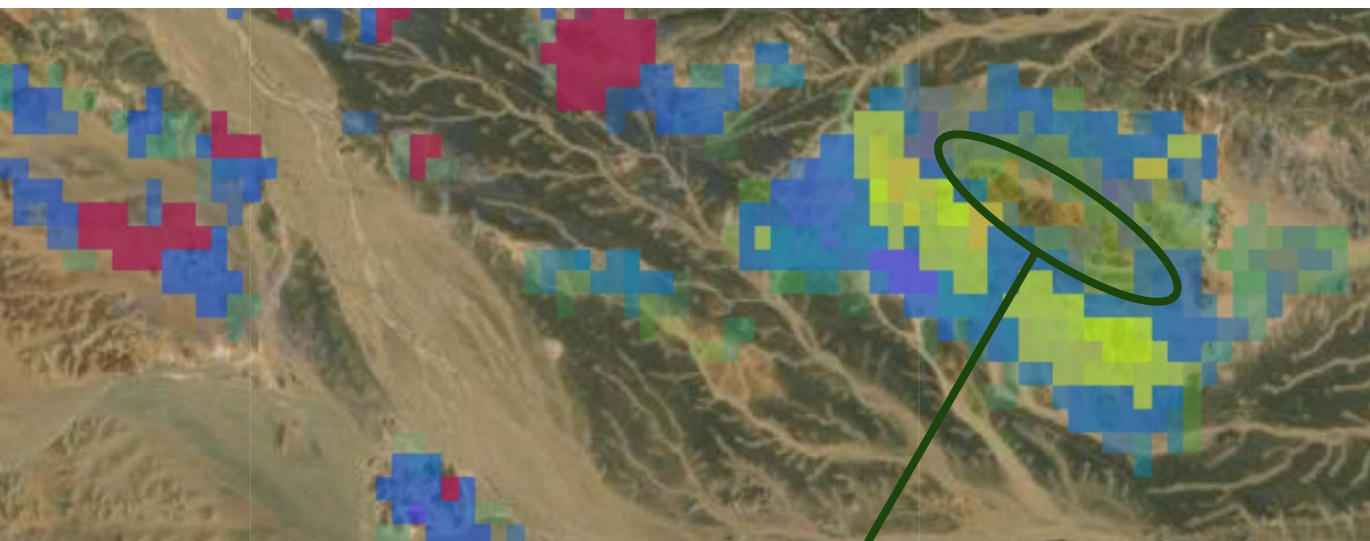
Propylitic Barren, except for subepither- mal veins	Marginal parts of systems, below lithocaps (ubiquitous)	Chlorite, epidote, albite, carbonate	Actinolite, hematite, magnetite
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Chlorite-sericite (sericite-clay-chlorite) Common ore contributor	Upper parts of porphyry Cu core zones (common, particularly in Au- rich deposits)	Chlorite, sericite/illite, hematite (martite, specularite)	Carbonate, epidote, smectite
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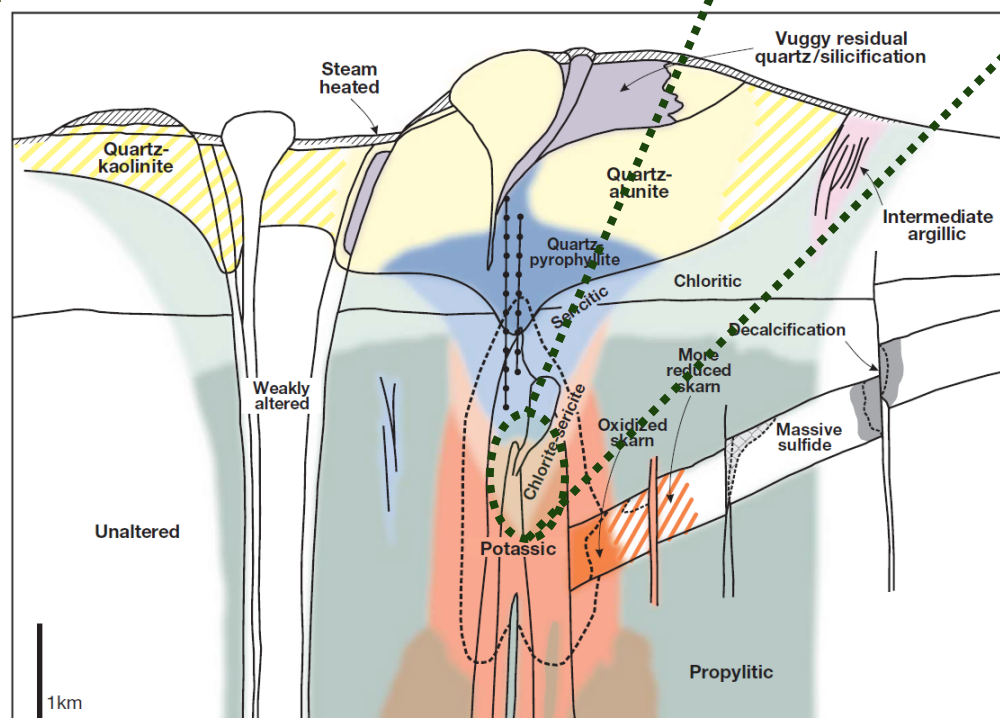
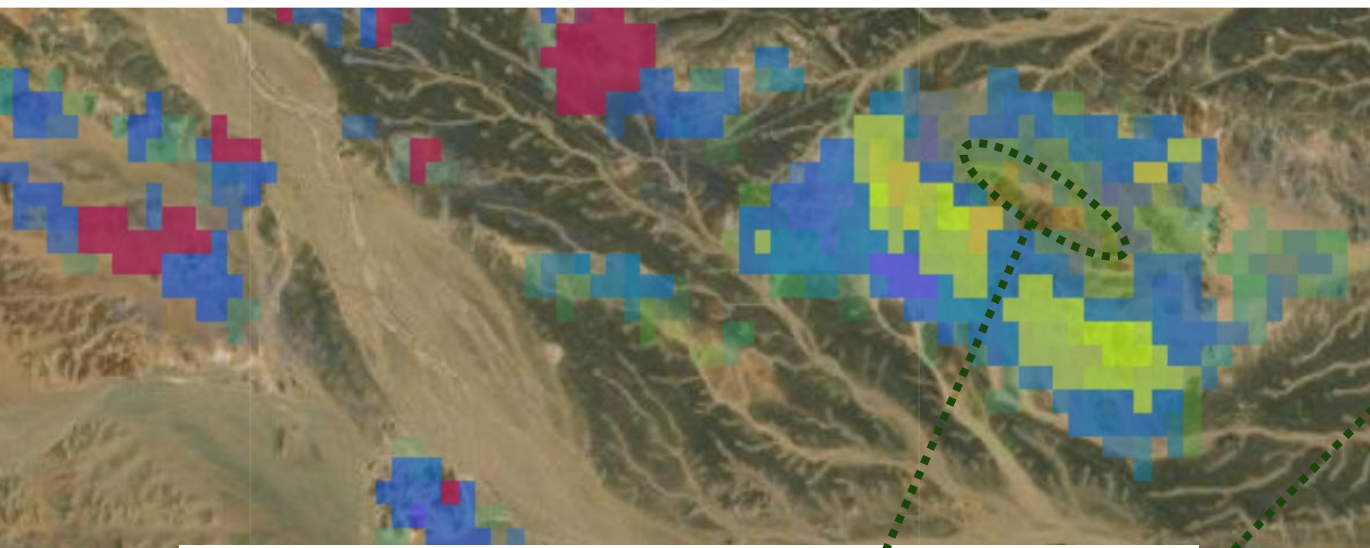
Sericitic (phyllic) Commonly barren, but may constitute ore	Upper parts of porphyry Cu deposits (ubiquitous, except with alkaline intrusions)	Quartz-sericite	Pyrophyllite carbonate, tourmaline, specularite
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Advanced argillic (secondary quartzite in Russian terminology) Locally constitutes ore in lithocaps and their roots	Above porphyry Cu deposits, constitutes lithocaps (common)	Quartz (partly residual, vuggy), alumite, ⁴ pyrophyllite,	Diaspore, andalusite, zunyite, corundum, dumortierite, topaz, specularite
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Alteration type ² (alternative name)	Position in system (abundance)	Key minerals	Possible ancillary minerals
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Potassic (K-silicate) Main ore contributor	Core zones of porphyry Cu deposits (ubiquitous)	Biotite, K-feldspar	Actinolite, epidote, sericite, andalusite, albite, carbonate, tourmaline, magnetite
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Chlorite-sericite (sericite-clay-chlorite) Common ore contributor	Upper parts of porphyry Cu core zones (common, particularly in Au- rich deposits)	Chlorite, sericite/illite, hematite (martite, specularite)	Carbonate, epidote, smectite
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Sericite (Muscovite) drop – converted to biotite			

Propylitic Barren, except for subepithemal veins	Marginal parts of systems, below lithocaps (ubiquitous)	Chlorite, epidote, albite, carbonate	Actinolite, hematite, magnetite
Chlorite-sericite (sericite-clay-chlorite) Common ore contributor	Upper parts of porphyry Cu core zones (common, particularly in Au- rich deposits)	Chlorite, sericite/illite, hematite (martite, specularite)	Carbonate, epidote, smectite
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The **EMIT (Earth Surface Mineral Dust Source Investigation)** payload aboard the International Space Station captures hyperspectral imagery across the visible and shortwave infrared (VSWIR) spectrum. The raw images are compiled into spatial-spectral "cubes". The key sensor and image parameters for EMIT data include:

Spectral Parameters

Wavelength Range: 380 to 2,500 nm

Number of Bands: 285 contiguous spectral bands

Spectral Sampling: ~7.4 nm per band

Spectral Response (FWHM): 8.5 nm

Spatial & Geometric Parameters

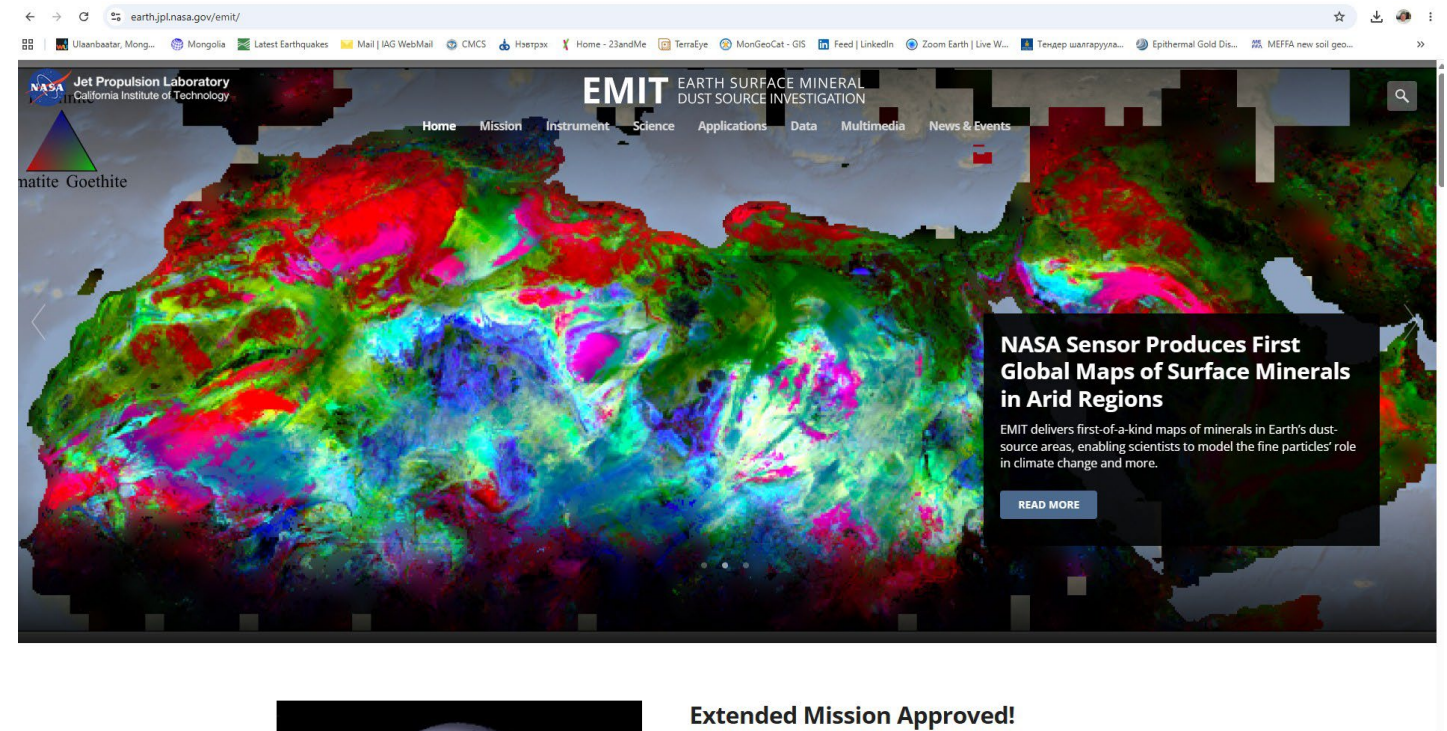
Ground Sampling Distance (GSD): ~60 meters

Swath Width: ~75 kilometers (nominal at the equator)

Cross-Track Pixels: 1,240 cross-track spatial elements

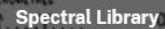
Along-Track Extent: At least 1,280 rows per scene, occasionally up to 2,559 rows

Geometric Correction: Orthorectified to a North-up WGS-84 projection



Extended Mission Approved!

<https://earth.jpl.nasa.gov/emit/>



MAPS

Researchers at the USGS Spectroscopy Lab are studying and applying methods for identifying and mapping materials through spectroscopic remote sensing (called imaging spectroscopy, hyperspectral imaging, imaging spectrometry, ultraspectral imaging, etc), on the earth and throughout the solar system using laboratory, field, airborne and spacecraft spectrometers.

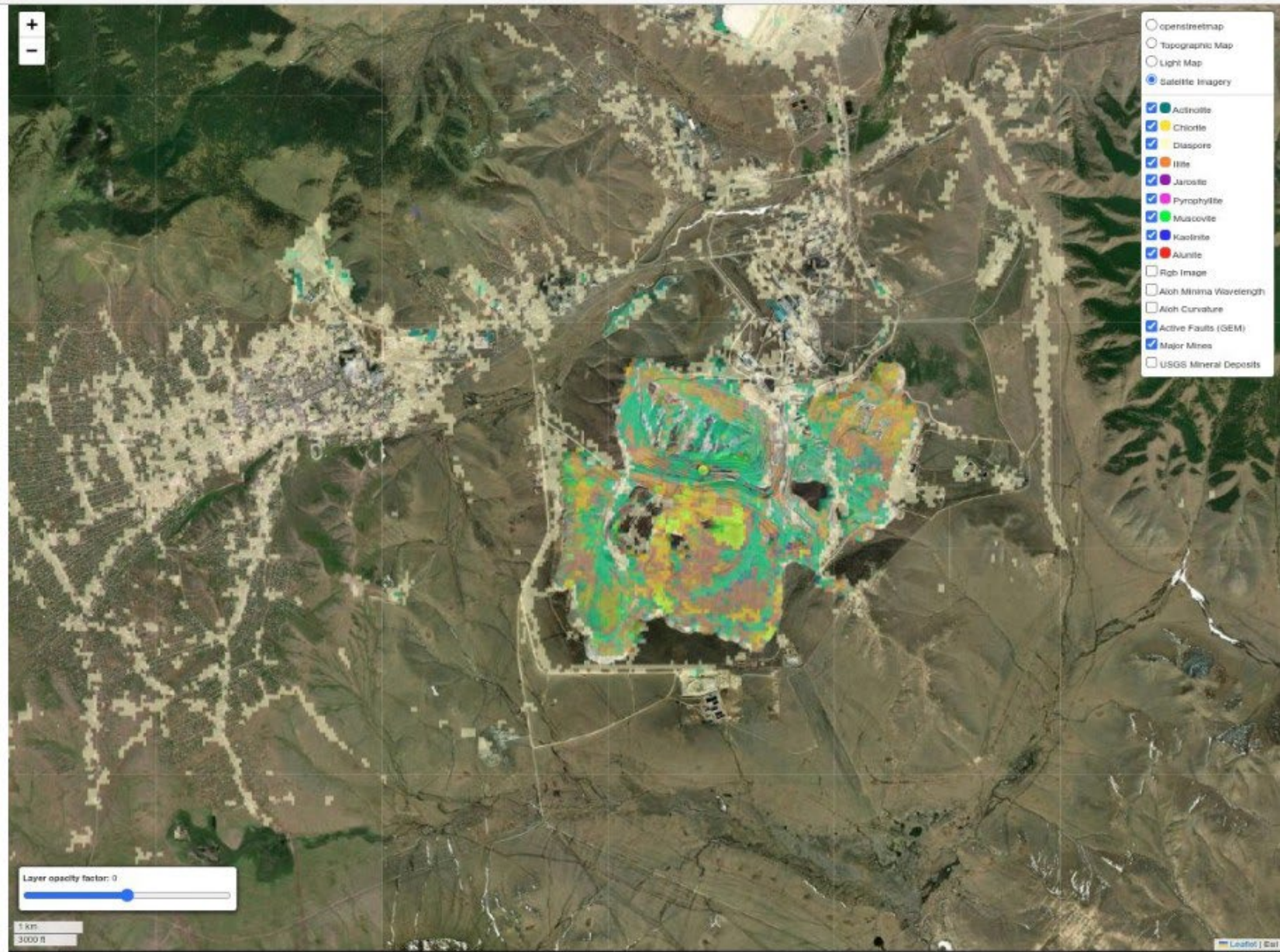
<https://www.usgs.gov/labs/spectroscopy-lab/science/spectral-library>

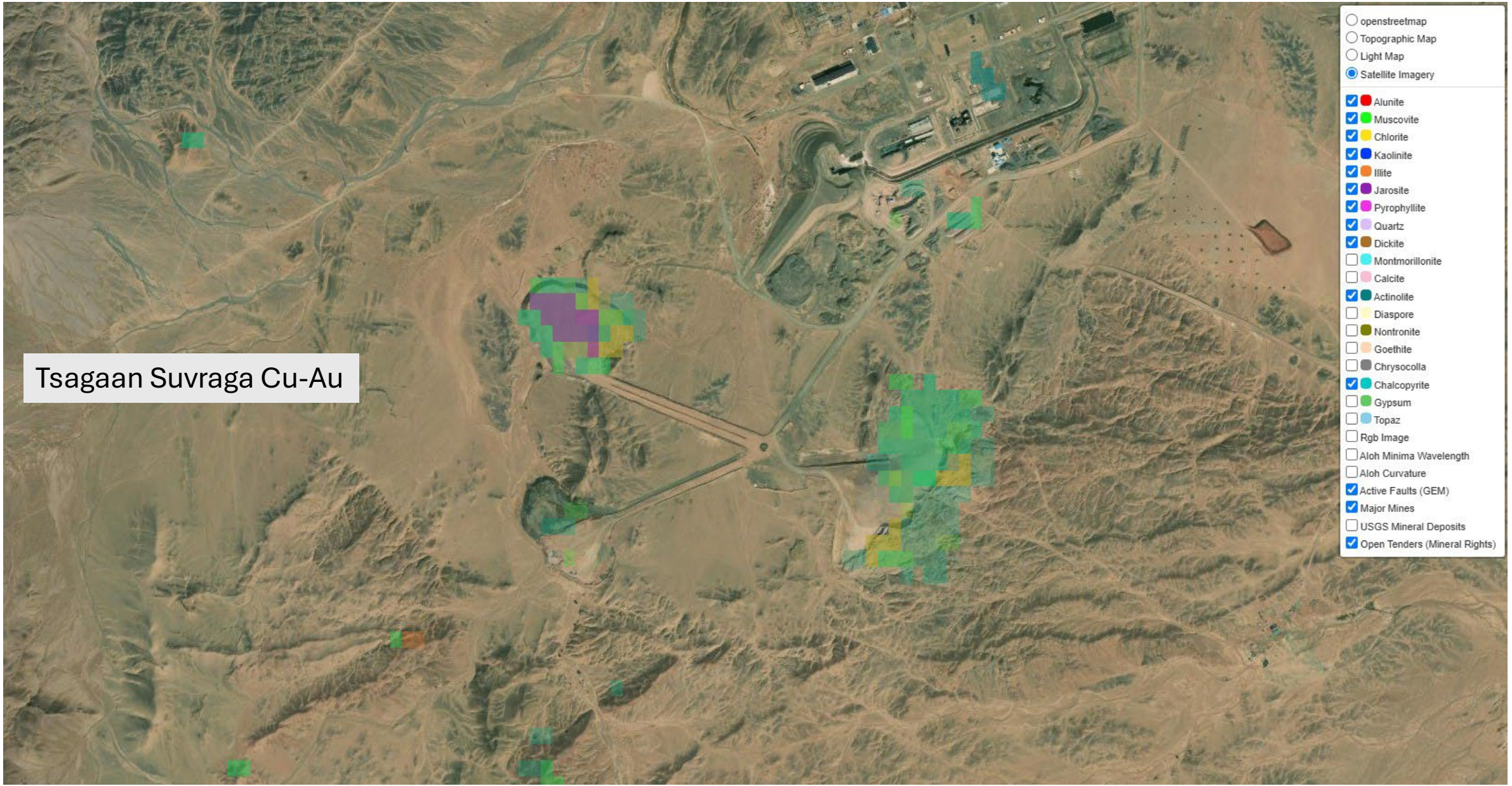
Erdenet

Commodity: Copper-Molybdenum

Status: Operating

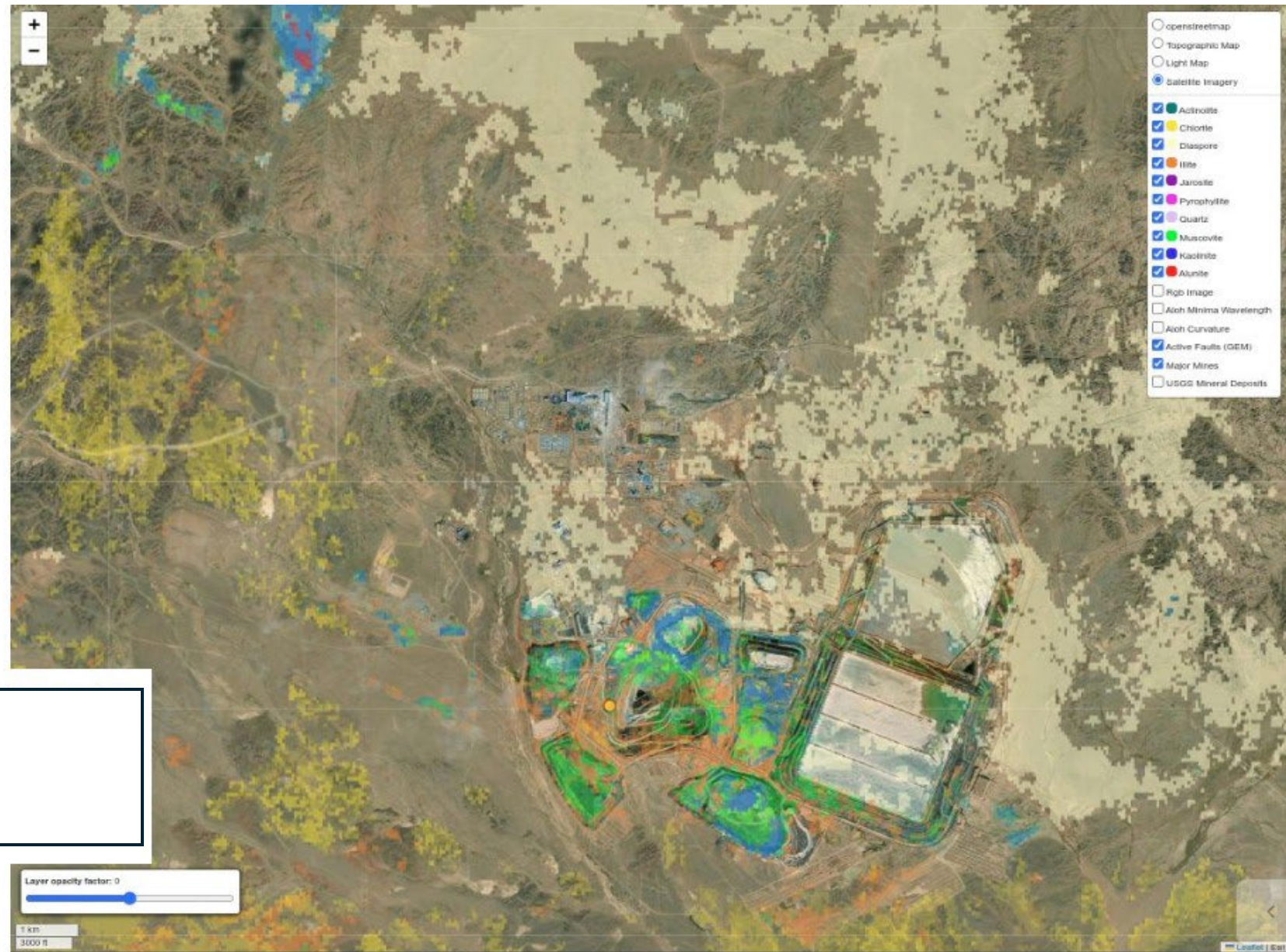
Operator: Erdenet Mining Corp



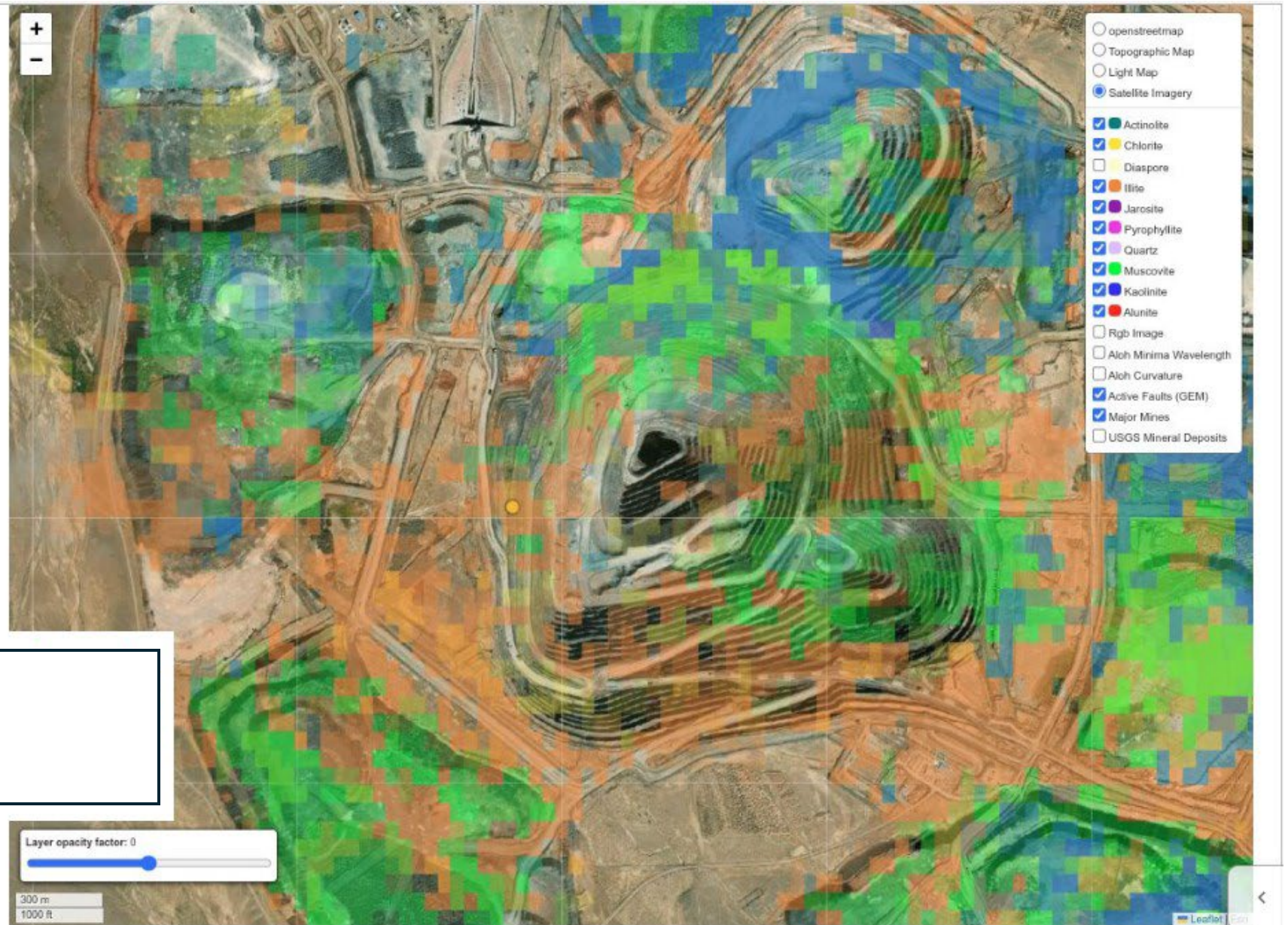


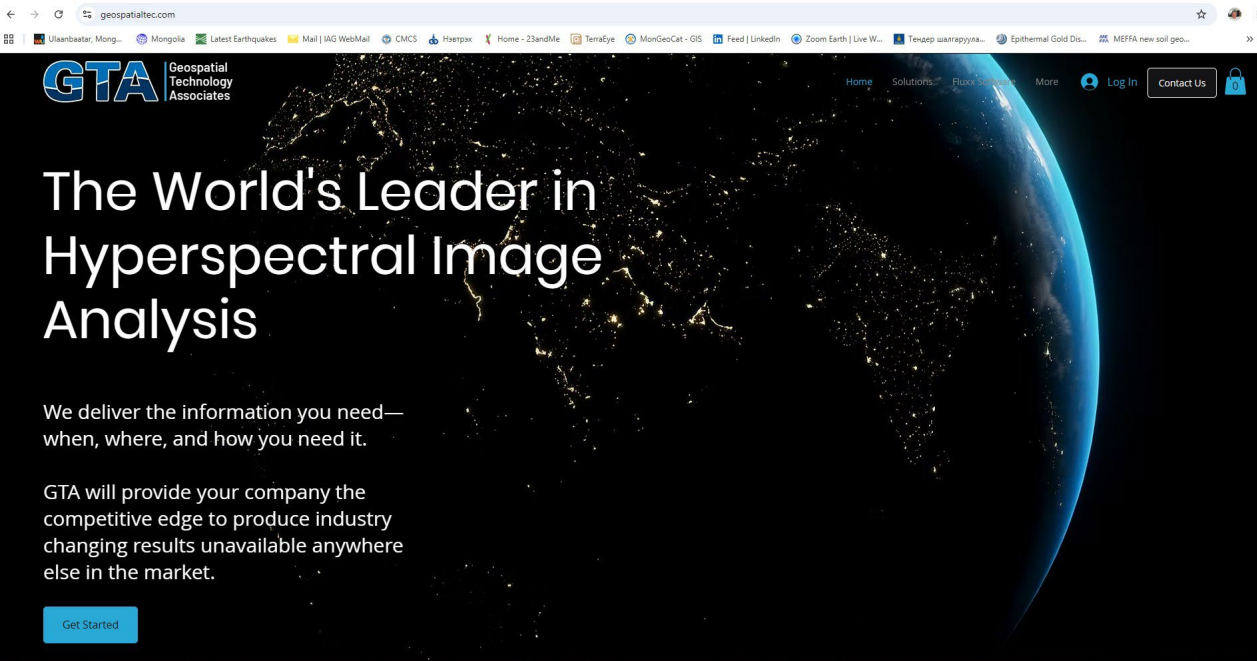
- ☐ openstreetmap
 - ☐ Topographic Map
 - ☐ Light Map
 - ☒ Satellite Imagery
-
- ☒ Alunite
 - ☒ Muscovite
 - ☒ Chlorite
 - ☒ Kaolinite
 - ☒ Illite
 - ☒ Jarosite
 - ☒ Pyrophyllite
 - ☒ Quartz
 - ☒ Dickite
 - ☐ Montmorillonite
 - ☐ Calcite
 - ☒ Actinolite
 - ☐ Diaspore
 - ☐ Nontronite
 - ☐ Goethite
 - ☐ Chrysocolla
 - ☒ Chalcopyrite
 - ☐ Gypsum
 - ☐ Topaz
 - ☐ Rgb Image
 - ☐ Aloh Minima Wavelength
 - ☐ Aloh Curvature
 - ☒ Active Faults (GEM)
 - ☒ Major Mines
 - ☐ USGS Mineral Deposits
 - ☒ Open Tenders (Mineral Rights)

name: Oyu Tolgoi
commodity: Copper-Gold
status: Operating
operator: Rio Tinto

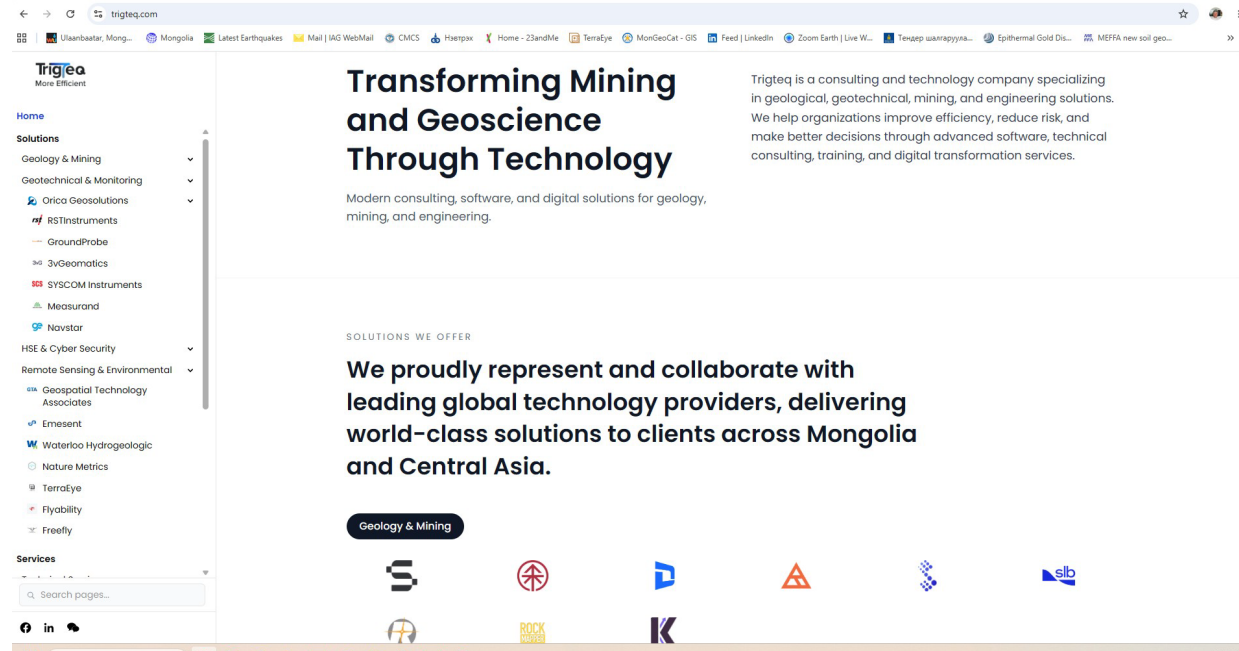


name: Oyu Tolgoi
commodity: Copper-Gold
status: Operating
operator: Rio Tinto





<https://www.geospatialtec.com/>



<https://trigteq.com/>

GTA

TrigTeeq
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